LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 12 Test: Magnetic Effects of Electric Current

Chapter / Topic: Magnetic Effects of Electric Current

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.

SECTION A

6 marks

Choose the correct option.

Q1. Which of the following correctly describes the magnetic field near a long, straight, current-carrying wire? **[1]**

(a) Straight lines perpendicular to the wire
(b) Straight lines parallel to the wire
(c) Concentric circles centred on the wire
(d) Radial lines starting from the wire

Q2. The magnetic field inside a long, straight, current-carrying solenoid: **[1]**

(a) is zero
(b) decreases towards its ends
(c) increases towards its ends
(d) is the same at all points

Q3. A positively charged particle (alpha particle) moving towards the west is deflected towards the north by a magnetic field. The direction of the magnetic field is: **[1]**

(a) towards the south
(b) upward
(c) towards the east
(d) downward

Q4. Magnetic field lines: **[1]**

(a) can intersect each other
(b) intersect only at the poles
(c) never intersect each other
(d) intersect at neutral points

Q5. Which device protects a household circuit from short-circuiting and overloading? **[1]**

(a) Ammeter
(b) Voltmeter
(c) Rheostat
(d) Fuse

Q6. **Assertion (A):** Two magnetic field lines never cross each other. **Reason (R):** If they crossed, a compass needle placed at the crossing point would have to point in two directions at once, which is not possible. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. State Fleming's left-hand rule. **[2]**

OR

Give two properties of magnetic field lines.

Q8. What is the function of the earth wire? Why should metallic appliances be earthed? **[2]**

SECTION C

6 marks

Q9. Answer the following about a solenoid: **[3]**

(i) What is a solenoid? **[1]**

(ii) How does the magnetic field of a current-carrying solenoid compare with that of a bar magnet? **[1]**

(iii) How can a solenoid be used to make an electromagnet? **[1]**

Q10. Answer the following: **[3]**

(i) State the right-hand thumb rule. **[1]**

(ii) A current flows from east to west in a horizontal power line. What is the direction of the magnetic field at a point directly below it, and at a point directly above it? **[2]**

OR

On which three factors does the strength of the magnetic field inside a current-carrying solenoid depend?

SECTION D

5 marks

Q11. Answer the following about domestic electric circuits: **[5]**

(i) Name the three wires used in domestic wiring, with the colour of their insulation. What is the potential difference between the live and neutral wires in India? **[3]**

(ii) What is meant by short-circuiting and overloading? **[2]**

OR

(a) A current-carrying straight conductor is placed in a magnetic field, perpendicular to it. How does the force on it change if (i) the current is increased, (ii) the direction of the current is reversed, (iii) the conductor is placed parallel to the field? [3]
(b) Name two devices that use current-carrying conductors and magnetic fields. [2]

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. Rohan's house has two separate circuits: one rated 15 A for high-power appliances like the geyser and AC, and one rated 5 A for fans and bulbs. One day, an electric heater, an iron and the AC were all switched on together from the same circuit, and the MCB tripped. **[4]**

(i) Why are separate circuits with different current ratings used? **[1]**

(ii) Why did the MCB trip? **[1]**

(iii) Calculate the current drawn by a 2 kW heater on a 220 V supply. Can it be used safely on the 5 A circuit? **[2]**

~ All the best! ~